Zaiqian "Ting" Chen

zc2666@columbia.edu
tingtang2.github.io
tingtang2

Education

2024–Present **PhD, Statistics**, *Columbia University*, New York, NY.

2022–2024 **MS, Computer Science**, *Columbia University*, New York, NY, GPA: *3.99/4.00*.

2017–2021 **BS, Computer Science (honors), Mathematical Economics, Summa Cum Laude**, *University of Richmond*, Richmond, VA, GPA: *3.94/4.00*.

Fall 2019 **Visiting Student**, *University of Edinburgh*, Edinburgh, Scotland, GPA: *4.0/4.0* (converted).

Peer Reviewed Publications

ML4LMS Workshop @ ICML 2024 S. Chang, **Z. Chen**, B. Dumitrascu, D. A. Knowles, *CellFlows: Inferring Splicing Kinetics from Latent and Mechanistic Cellular Dynamics* in ICML'24 Workshop ML for Life and Material Science: From Theory to Industry Applications. 2024.

EMNLP 2022 **Z. Chen** D. Verdi do Amarante, J. Donaldson, Y. Jo and J. Park, *Argument Mining for Review Helpfulness Prediction* in Proceedings of the Empirical Methods on Natural Language Processing (EMNLP). 2022.

DSAA 2021 **Z. Chen** and J. Park, *Analyzing Cultural Assimilation through the Lens of Yelp Restaurant Reviews* at the 8th IEEE International Conference on Data Science and Advanced Analytics (DSAA 2021).

Research Experience

2025-Present **PhD Student**, *Department of Statistics: Professor John Cunningham, Department of Electrical Engineering: Professor Micah Goldblum*, Columbia University.
- Developing computation-aware methods for high dimensional and latent space Bayesian optimization
- Analyzing LLM training dynamics and identifying causes behind pretraining loss spikes

2022–2024 **Graduate Research Assistant**, *Department of Computer Science: Professor David Knowles*, Columbia University, New York Genome Center.
- Developed deep kernel learning-based active learning methods to efficiently learn splicing outcomes from DNA sequences
- Trained VAEs and latent neural ODEs to infer cellular dynamics from scRNA-seq data

2020–2021 **Honors Thesis**, *Department of Computer Science: Professor Joonsuk Park*, University of Richmond.
- Implemented transfer learning methods for graph-based argument mining to improve helpfulness prediction of Amazon product reviews

Spring, Summer 2020 **Research Assistant**, *Department of Computer Science: Professor Joonsuk Park*, University of Richmond.
- Developed probabilistic graphical models and inference algorithms to perform topic modeling on online review data for cultural analysis

Work Experience/Service

Spring 2025 **Teaching Assistant - Statistical Machine Learning**, *Department of Statistics*, Columbia University.

Fall 2024, 2025 **Teaching Assistant - Advanced Machine Learning**, *Department of Statistics*, Columbia University.

2021-2024 **Machine Learning Software Engineer**, *CarMax*.
- Built personalized recommendation systems and customer segmentation systems for search engine optimization
- Built microservices, Python libraries, and CI/CD pipelines in order to support data sourcing and model orchestration
- Trained, deployed, and served model predictions through AzureML, Apache Spark, and Databricks

Fall 2024 **Reviewer**, *ICML Workshop - CODEML: Championing Open-source DEvelopment in Machine Learning*.

Fall 2023 **Reviewer**, *Nature Machine Intelligence*.

Spring 2019, Spring 2021 **Grading Assistant - Data Structures, Computer Systems and Organization**, *Department of Computer Science*, University of Richmond.

Spring 2019 **Grading Assistant - Linear Algebra**, *Department of Mathematics*, University of Richmond.

Technical Skills

Relevant Coursework Bayesian Optimization, Foundations of Graphical Models, Machine Learning in Genomics, Deep and Generalized Probabilistic Models, Neural Networks and Deep Learning, Geometric Data Analysis, Interpretable Machine Learning for Inverse Problems in Biology

Software PyTorch, JAX, LaTeX, Apache Spark, Python, R, C++, Microsoft Azure Cloud Services, Google Cloud Platform (GCP), Azure Cosmos DB, Azure DevOps, Azure ML

Honors

2021 **Phi Beta Kappa**, University of Richmond

2017–2021 **University of Richmond Science Scholar Award:** Full tuition merit scholarship for 4 years

2017–2021 **National Merit Scholarship**

2017 **National AP Scholar**

2016 **Most Interesting Hack** at Vassar College Hackathon